REGIME.fi

AUTONOMOUS TRADING PROTOCOL

Whitepaper v1.0 — April 2026

01 — Abstract

An Autonomous, AI-Evolved Trading Fund

REGIME.fi is an autonomous trading fund built on Arbitrum. It executes systematic mean-reversion strategies across decentralized perpetual exchanges, generating yield from short-term market inefficiencies rather than token emissions or liquidity provision.

What makes REGIME.fi different is how it evolves. Every night, an AI system runs dozens of experiments against historical market data — testing new hypotheses, tuning parameters, and measuring results with scientific rigor. The strategies that trade your capital are not static. They improve continuously, guided by data rather than intuition.

All fund activity is fully transparent and verifiable on-chain. Every trade, every position, every NAV update is visible in real time on the public dashboard. Stakers deposit USDC and receive shares priced at the fund's current net asset value. There are no management fees — the team earns only when the fund profits.

24/7	Multi	AI	100%
AUTONOMOUS	ASSET CLASS	EVOLVED	ON-CHAIN

02 — The Problem

DeFi Yield Is Broken

Generating sustainable yield in DeFi remains one of the hardest problems in crypto. The most common approaches all carry fundamental flaws that transfer risk to the user while obscuring it behind attractive APY numbers.

Liquidity Provision

Automated market makers promise passive yield from trading fees, but impermanent loss quietly erodes principal. In volatile markets — which is most of the time in crypto — liquidity providers often end up worse off than simply holding the underlying assets. The math works against you precisely when you need it most.

Yield Farming

Token emission programs create the illusion of high yields by printing governance tokens. These programs are structurally unsustainable — they dilute token holders to subsidize temporary TVL. When emissions slow or stop, capital leaves and token prices collapse. The yield was never real; it was a transfer from future token holders to current ones.

Active Fund Management

On-chain funds managed by discretionary traders suffer from the same problems as traditional hedge funds: opaque decision-making, emotional trading, and misaligned incentives. Most charge management fees regardless of performance, meaning the team profits even when stakers lose money. Strategy logic is hidden, positions are obscured, and there is no way for stakers to verify that the fund is operating as promised.

A Better Approach

Sustainable yield requires a fundamentally different model — one that generates returns from genuine market inefficiencies, operates with full transparency, removes human emotion from execution, and aligns the team's incentives directly with fund performance.

> REGIME.fi was built to solve these problems: systematic trading that captures real edge, AI-driven evolution that keeps the strategy sharp, and on-chain architecture that makes every action verifiable.

Systematic Trading, Evolved by AI

REGIME.fi is an autonomous trading fund that executes mean-reversion strategies on decentralized perpetual exchanges. The fund trades across multiple venues and asset classes on Arbitrum — including crypto, forex, commodities, and indices — capturing short-term price dislocations that reliably snap back to equilibrium.

Mean reversion is one of the most well-documented phenomena in financial markets. When prices move too far, too fast, they tend to revert. REGIME.fi identifies these overextensions systematically using multiple confirming signals, then executes trades with precise risk controls. No human decides when to trade. No human decides how much to risk. The system follows its rules.

The fund is built on three pillars:

THE THREE PILLARS

Systematic Strategy

Rule-based mean-reversion trading with multi-signal confirmation and tiered conviction sizing. No discretionary decisions.

AI-Driven Evolution

Nightly AI experiments continuously optimize parameters and test hypotheses against historical data. The strategy improves every day.

On-Chain Transparency

Every trade, position, and NAV update is verifiable on-chain and visible on the public dashboard in real time.

Multi-Venue, Multi-Asset

The fund trades on multiple decentralized perpetual exchanges on Arbitrum, currently including gTrade (Gains Network) and Ostium. This multi-venue approach provides access to a broad range of asset classes:

Crypto	Forex	Commodities	Indices
BTC, ETH & MORE	MAJOR & MINOR PAIRS	GOLD, OIL & MORE	S&P 500, NASDAQ

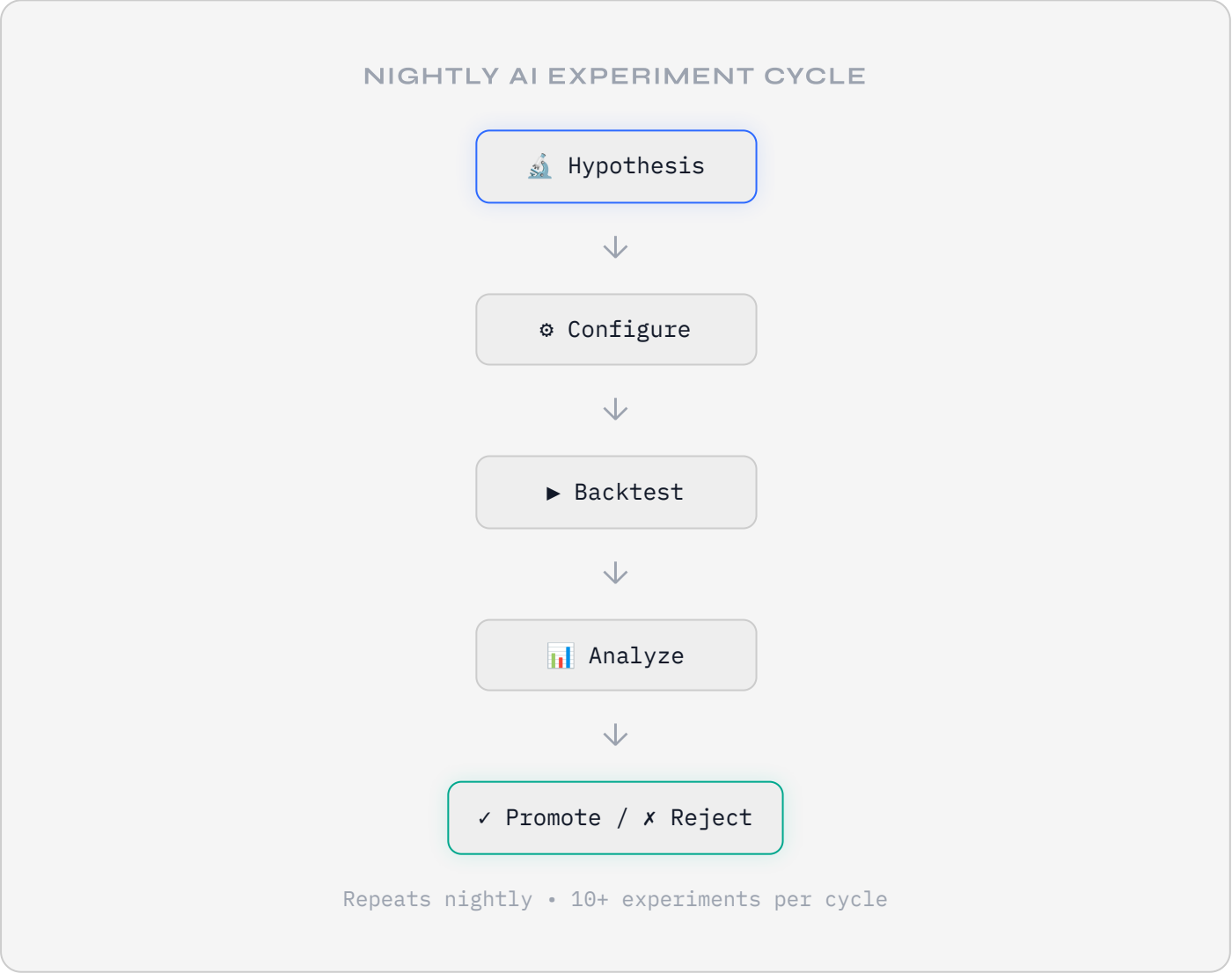
Each venue has its own fee structure, execution mechanics, and price feeds. The fund's fee model accurately simulates each venue's costs, ensuring that only trades with a genuine edge after fees are executed.

04 — AI-Driven Strategy Evolution

A Strategy That Improves Every Night

Most trading systems are designed once and deployed. Over time, market conditions shift, edge decays, and what once worked stops working. The traditional response is manual intervention — a trader reviews results, makes adjustments based on intuition, and hopes the changes help. This is slow, subjective, and prone to cognitive bias.

REGIME.fi takes a fundamentally different approach. Every night, an AI system — powered by Claude, Anthropic's most capable model — runs a structured experimentation cycle against the fund's historical trading data. This isn't a black-box optimizer searching for overfitted parameters. It's a hypothesis-driven research process:



How It Works

Each night, the AI reviews the fund's recent trading performance and identifies areas for potential improvement. It formulates specific, testable hypotheses — for example, whether adjusting a risk parameter would reduce drawdowns without sacrificing returns, or whether a new market filter could improve signal quality during trending conditions.

For each hypothesis, the AI configures a backtest: a replay of historical market data through the strategy engine with the proposed changes applied. The backtest uses the same venue-accurate fee models as live trading, ensuring that results reflect real-world economics. Spreads, rollover fees, funding rates, and execution costs are all accounted for.

After each backtest completes, the AI analyzes the results against the fund's standard evaluation framework — a consistent set of metrics including profit factor, win rate, drawdown, fee impact, and trade distribution. Changes are only promoted to the live strategy if they demonstrate clear, measurable improvement without introducing new risks.

> This process has produced over 15 documented iterations of strategy improvement since inception. Each iteration builds on the last, creating a compound effect where small, validated improvements accumulate into significant performance gains over time.

The Backtesting Framework

Rigorous backtesting is the foundation of trustworthy strategy development. The fund's backtesting engine replays historical market data through the complete strategy pipeline — from raw price feeds through signal generation, conviction scoring, position sizing, entry, exit, and settlement — exactly as it runs in production.

Key properties of the backtesting framework:

- **Venue-accurate fee models** — each exchange's specific fee structure is simulated precisely, including trading fees, spread costs, and rollover charges
- **No lookahead bias** — the engine processes candles sequentially, making decisions only on data available at the time
- **Consistent evaluation metrics** — every experiment is measured against the same standardized framework, making comparisons meaningful
- **Parameter sweep support** — systematic exploration of parameter spaces to find robust values rather than fragile optima

Why AI-Driven Evolution Matters

Markets are not static. Volatility regimes change, correlations shift, and new participants alter market microstructure. A strategy that works today may underperform tomorrow — not because it was poorly designed, but because conditions changed.

Continuous AI-driven experimentation addresses this directly. Rather than waiting for performance degradation to trigger manual review, the system proactively searches for improvements every night. This creates a feedback loop where the strategy is always adapting, always testing, and always evolving — guided by data rather than human intuition or emotion.

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05 — On-Chain Architecture

Smart Contracts on Arbitrum

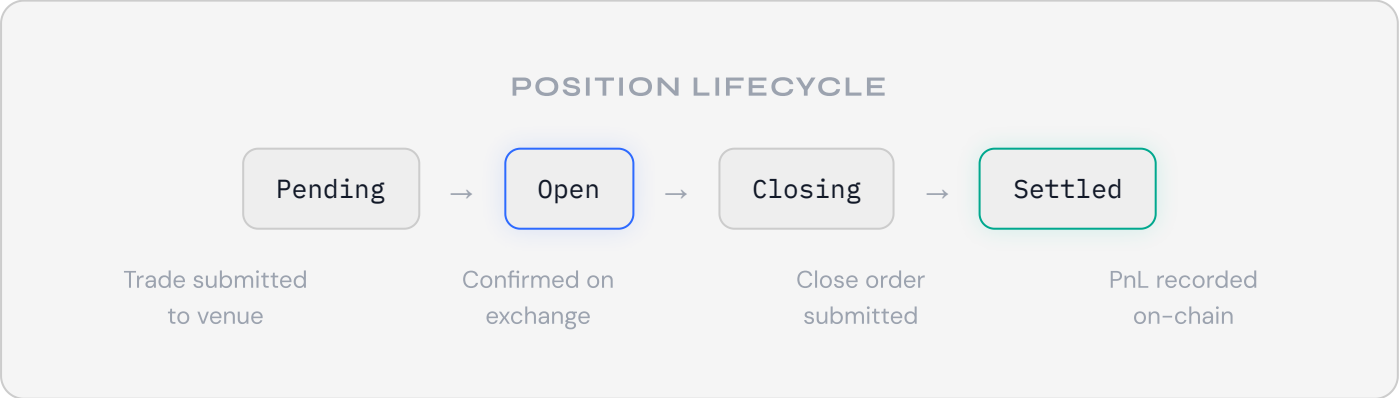
REGIME.fi's fund operations are managed by smart contracts deployed on Arbitrum, an Ethereum Layer 2 network that provides low transaction costs and fast settlement while inheriting Ethereum's security guarantees.

The on-chain architecture consists of two primary contracts:

RegimeStaking

The core fund contract. RegimeStaking manages staker deposits, share issuance, position execution, and withdrawal processing. All fund capital flows through this contract.

- **Share-based NAV system** — stakers receive shares proportional to their deposit at the current net asset value. As the fund trades profitably, NAV increases, and each share becomes worth more USDC.
- **Epoch-based withdrawals** — withdrawal requests are processed in daily settlement cycles (epochs), ensuring fair pricing for both departing and remaining stakers.
- **Position lifecycle management** — every trade follows a strict on-chain lifecycle with safety checks at each stage.
- **Safety caps** — no single trade can exceed 10% of the fund's total value, and total active exposure is capped at 50%. These limits are enforced at the contract level and cannot be overridden.



RegimeTreasury

A separate contract that receives the team's share of trading profits. The treasury splits commissions evenly among team members, who can claim their share on demand. The contract enforces fair distribution — the owner cannot drain team commissions, and the rescue function explicitly protects allocated funds.

Security by Design

The contracts follow a principle of minimal approval: instead of granting blanket USDC spending authority, each operation approves only the exact amount required. Combined with the position safety caps, this creates multiple layers of protection for staker funds at the smart contract level.

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06 — Staking & NAV

How Staking Works

Staking into REGIME.fi is designed to be simple and fair. You deposit USDC, receive shares representing your ownership of the fund, and your shares appreciate in value as the fund trades profitably.

Depositing

When you deposit USDC into the fund, you receive shares at the current NAV (net asset value) price. The NAV starts at $1.00 per share and moves based on the fund's trading performance:

NAV Formula:

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NAV = (Total USDC + Active Positions - Withdrawals Owed) / Total Shares
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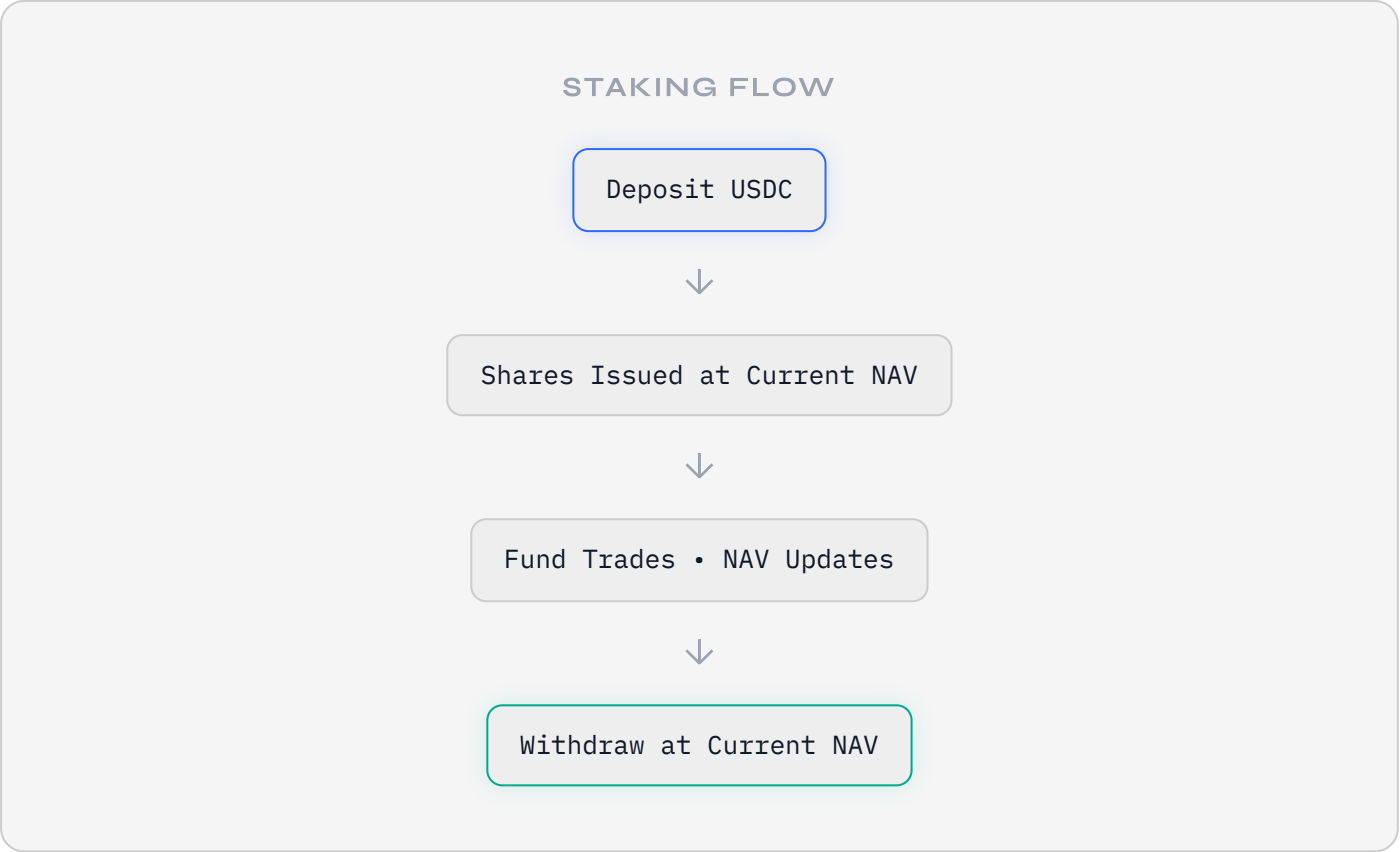
For example, if the NAV is $1.15 and you deposit 1,000 USDC, you receive approximately 869.57 shares. If the fund continues to perform and NAV rises to $1.30, your shares are now worth 1,130.43 USDC — a 13% gain.



Withdrawing

Withdrawals use an epoch-based settlement process. When you request a withdrawal, your shares are queued for the next settlement cycle. At the end of the epoch, all pending withdrawals are processed at the finalization NAV — the fund's value at the moment of settlement.

This design is intentional: pricing withdrawals at finalization NAV (rather than request NAV) prevents front-running and ensures fair treatment for stakers who remain in the fund. You always receive the true value of your shares at the time they are settled.

Fair Pricing Guarantees

- **No entry/exit fees** — you pay no fees to deposit or withdraw
- **Epoch finalization pricing** — withdrawals are valued at settlement time, preventing manipulation
- **Pro-rata share distribution** — shares represent proportional ownership, not a fixed USDC amount
- **Transparent NAV calculation** — the formula accounts for active positions and pending obligations, visible on-chain

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Every Trade Is Sized to Lose

The fund's risk management philosophy starts from a simple premise: every trade might be a loser. Rather than optimizing for large wins, the system is designed so that losing trades have minimal impact while winning trades accumulate steadily over time. Edge comes from consistency, not from any single position.

Tiered Conviction Sizing

Not all trading signals are created equal. The fund uses a tiered conviction system that allocates more capital to higher-confidence setups and less to weaker signals. Each tier has its own leverage limits and risk allocation, ensuring that position sizing is always proportional to signal strength.

TIER	SIGNAL QUALITY	RISK ALLOCATION	PURPOSE
High Conviction	Strong overextension + confirming signals	Largest	Primary alpha generation
Medium Conviction	Moderate overextension or partial confirmation	Moderate	Supplementary returns
Research	Weak signals, testing hypotheses	Minimal	Data collection for strategy improvement

Position Controls

- **Position limits** — maximum concurrent positions prevent overexposure to correlated moves
- **Priority eviction** — high-conviction signals can replace lower-conviction positions, ensuring capital is always allocated to the best opportunities
- **Cooldown periods** — after a loss, the system pauses before re-entering, preventing revenge trading and allowing conditions to reset
- **Maximum hold duration** — positions that haven't reached their target within a time window are closed, preventing capital from being tied up in stale trades

Fee-Aware Execution

Every potential trade is validated against the venue's actual fee structure before execution. If the expected round-trip fees (entry + exit) would consume too large a portion of the expected profit, the trade is skipped. This prevents the fund from taking trades where fees eat the edge — a common problem with high-frequency strategies on leveraged exchanges.

Trailing Stops

When a position reaches a meaningful profit target, a trailing stop activates to protect gains. Rather than waiting for either the take-profit or stop-loss, the trailing stop locks in a portion of the unrealized profit and follows the price as it continues to move favorably. This mechanism captures additional upside while limiting the risk of giving back gains.

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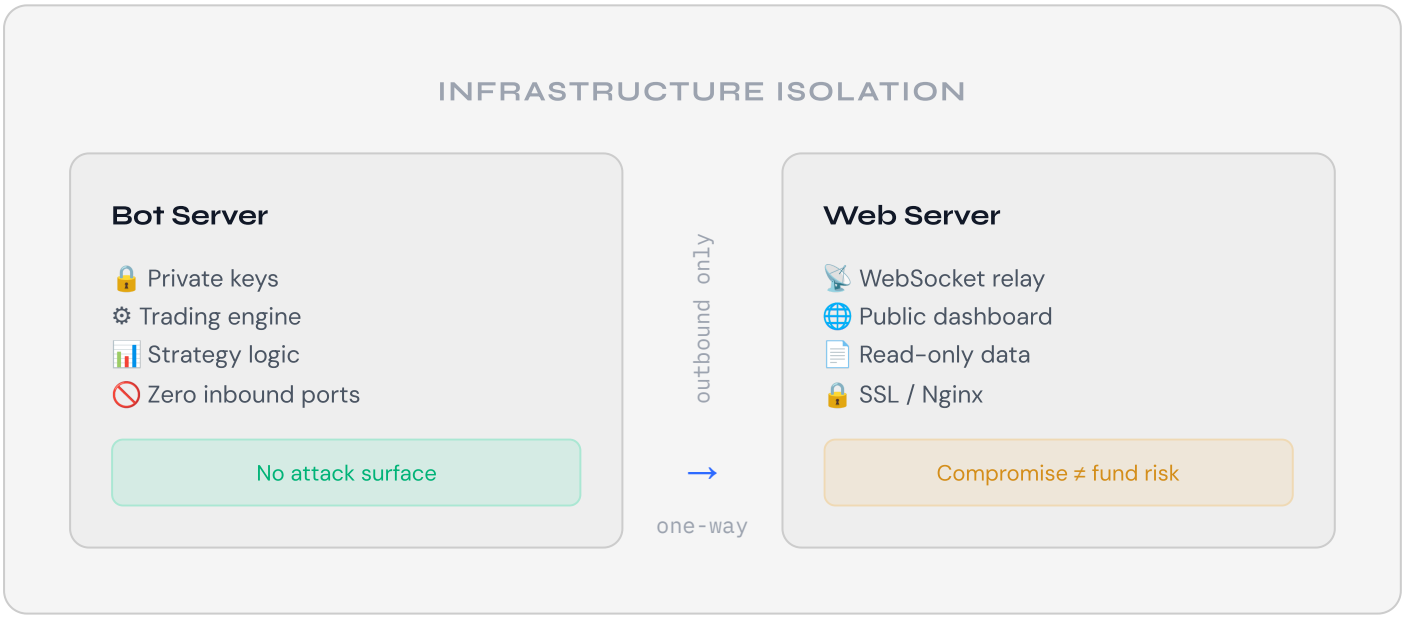
08 — Security Model

Zero-Trust Architecture

REGIME.fi's infrastructure is designed around a core principle: the trading bot — which manages private keys and executes trades with real capital — must have zero attack surface. This is achieved through strict server isolation and one-way data flow.



How Isolation Works

- **No shared code** — the bot and dashboard share zero dependencies, imports, or packages. Types are intentionally duplicated rather than shared, eliminating supply chain attack vectors between the systems.
- **No shared infrastructure** — separate servers, separate filesystems, separate processes. No database connections, no shared secrets beyond the relay authentication token.
- **Outbound-only communication** — the bot connects outbound to the relay. No ports are open on the bot server. An attacker who compromises the web server cannot initiate any connection to the bot.
- **One-way data flow** — data flows from bot to relay to dashboard. No messages flow back. The relay broadcasts trading data to connected dashboards but cannot send commands to the bot.

On-Chain Safety Layers

Beyond infrastructure isolation, the smart contracts enforce their own safety constraints:

- **Position caps** — single trade ≤10% of fund, total exposure ≤50% of fund
- **Exact approvals** — no blanket USDC spending authority; each operation approves only the precise amount needed
- **On-chain verification** — settlement prices and PnL are verified against the venue's on-chain records

> The security model ensures that even in a worst-case scenario — full compromise of the web server and dashboard — an attacker gains access to trading data but cannot execute trades, move funds, or send any command to the bot. The fund's capital remains secure.

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09 — Transparency

No Black Box

Transparency is not an afterthought — it is a core design principle. Every action the fund takes is visible in real time on the public dashboard at **regime.fi**.

The dashboard provides a live view of:

- **Active positions** — current trades with entry prices, direction, leverage, and unrealized PnL
- **Trade history** — every closed position with full details: entry, exit, fees, net profit or loss
- **NAV history** — the fund's net asset value over time, charted for easy visualization of performance trends
- **Portfolio metrics** — aggregate statistics including profit factor, win rate, and drawdown
- **Fund composition** — total staked value, share count, current NAV price

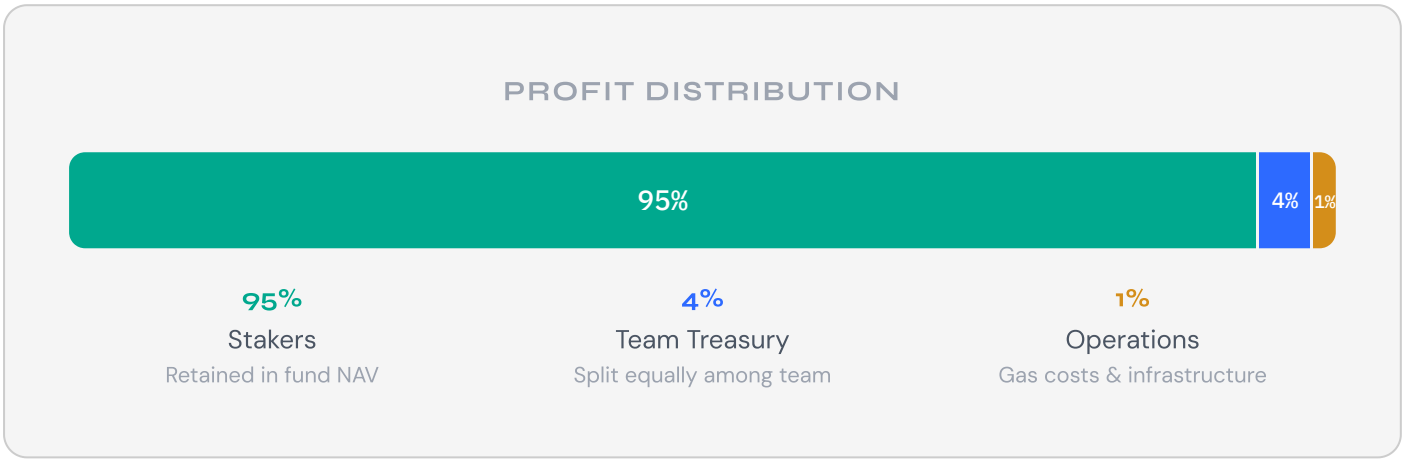
Data is delivered via WebSocket, meaning the dashboard updates in real time — not on a delay, not on a refresh cycle. When a trade opens or closes, you see it immediately.

> Traditional funds publish monthly or quarterly reports. REGIME.fi publishes every trade, every second. Stakers never have to wonder what the fund is doing with their capital.

10 — Fee Structure

Aligned Incentives

REGIME.fi charges no management fees, no deposit fees, and no withdrawal fees. The team earns only from a share of the fund's trading profits. If the fund doesn't make money, neither does the team.



This fee structure is enforced at the smart contract level. The 5% total fee is deducted only from profitable trades — when a position is settled with a gain, 1% goes to the operations wallet for gas costs and infrastructure, and 4% goes to the RegimeTreasury contract where it is split equally among team members.

Team members claim their share from the treasury on demand. The contract ensures that the owner cannot drain accumulated team commissions, and the rescue function explicitly protects allocated funds from being swept.

> **No profit = no fees.** The team only earns when stakers earn. This is the simplest and most honest incentive alignment possible.

11 — Roadmap

What's Next

REGIME.fi is a live, functioning system — not a concept or a promise. The roadmap below reflects the team's current priorities, presented honestly and without commitments to specific timelines.

Smart Contract Audit

Independent security audit of RegimeStaking and RegimeTreasury contracts before accepting external staker deposits.

Additional Venues

Integration with additional decentralized perpetual exchanges to expand the universe of tradeable pairs and capture more opportunities.

Strategy Diversification

Additional alpha sources beyond mean-reversion — expanding the strategy portfolio to capture edge from multiple independent signal types.

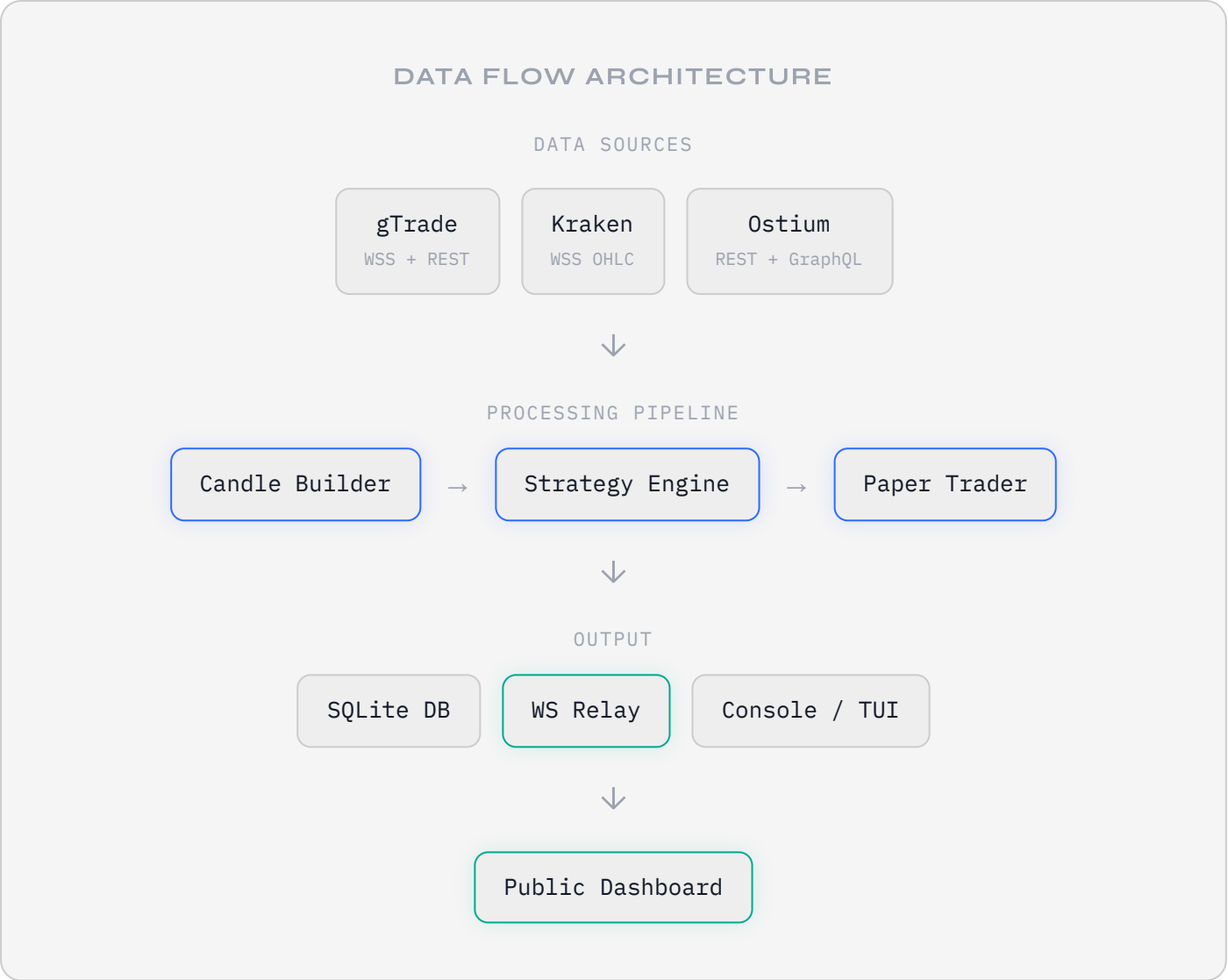
Governance

Evaluating governance structures that give stakers a voice in fund parameters and strategy direction, without compromising execution quality.

Appendix A

System Architecture

The REGIME.fi system consists of three isolated components: the trading bot, the WebSocket relay, and the public dashboard. Data flows in one direction only — from the bot to the relay to connected dashboards.



Trading Bot

The core engine collects real-time market data from multiple venues, normalizes it into standardized candles, runs it through the mean-reversion strategy engine, and executes trades via the paper trader (or live trader). Each strategy runs in isolation with its own configuration, state, and database records.

WebSocket Relay

A lightweight broadcast service that accepts an outbound connection from the bot and distributes trading data to connected dashboard clients. The relay authenticates the bot via a shared secret and serves as a one-way bridge between the private trading infrastructure and the public dashboard.

Database

The bot uses SQLite for persistent storage, with per-strategy isolation via strategy ID columns. Key tables include position records, trade signals, signal scorecards, fund share events, NAV history, and market snapshots. The database is local to the bot server and is never exposed to external systems.

Appendix B

Smart Contract Reference

Both contracts are deployed on Arbitrum One. Source code is available in the project's contracts repository.

Contract Addresses

CONTRACT	NETWORK	NOTES
RegimeStaking	Arbitrum One	Main fund contract
RegimeTreasury	Arbitrum One	Team commission distribution

Key External Dependencies

CONTRACT	ADDRESS	PURPOSE
USDC	`0xaf88...5831`	Fund denomination token
gTrade Diamond	`0xFF16...f169`	Gains Network trading
Ostium Trading	`0x6D0b...2411`	Ostium exchange

RegimeStaking — Key Functions

FUNCTION	ACCESS	DESCRIPTION
`stake(uint256)`	Public	Deposit USDC, receive shares at current NAV
`requestWithdrawal(uint256)`	Public	Queue shares for next epoch settlement
`claimWithdrawal()`	Public	Claim settled USDC after epoch finalization
`openPosition(...)`	Owner	Submit trade to venue with safety cap checks
`settlePosition(...)`	Owner	Record PnL and update NAV after trade closes
`finalizeEpoch()`	Owner	Process pending withdrawals at current NAV

Safety Invariants

- Single position collateral ≤ 10% of total fund value
- Sum of all active position collateral ≤ 50% of total fund value
- USDC approval per operation = exact amount needed (no blanket approvals)
- Withdrawal claims only available after epoch finalization
- Position status transitions are strictly ordered: Pending → Open → Closing → Settled

RegimeTreasury — Key Functions

FUNCTION	ACCESS	DESCRIPTION
`receiveCommission(uint256)`	Public	Accept USDC commission from staking contract
`claim()`	Member	Claim accumulated commission share
`addMember(address)`	Owner	Add a team member to commission split
`rescue(address, uint256)`	Owner	Recover stuck tokens (protects team commissions)

Appendix C

Venue Integration

The fund connects to multiple decentralized perpetual exchanges, each with its own API, fee model, and asset coverage. The strategy engine handles venue-specific differences transparently, presenting a unified interface to the trading logic.

gTrade (Gains Network)

A decentralized leveraged trading platform on Arbitrum supporting crypto, forex, and stock pairs. The fund connects via both WebSocket (real-time price updates) and REST API (trade execution and position management).

PROPERTY	DETAIL
Asset Classes	Crypto, forex, stocks
Data Feed	WebSocket (real-time) + REST
Fee Model	Opening/closing fees, spread, rollover, funding rate
Price Scaling	Prices scaled by 1e10

Ostium

A decentralized exchange focused on forex and commodities trading on Arbitrum. Provides broad coverage of traditional finance asset classes in a DeFi-native format.

PROPERTY	DETAIL
Asset Classes	Forex, commodities, indices, crypto, metals, energy
Pairs	58 pairs across 6 asset classes
Data Feed	REST polling + GraphQL subgraph
Fee Model	Dynamic fees based on OI skew and utilization

Fee Model Accuracy

Each venue integration includes a dedicated fee model that accurately calculates the total cost of each trade. The backtesting engine uses these same fee models, ensuring that backtest results reflect real trading economics. Fees accounted for include trading commissions, spread costs, rollover/funding charges, and venue-specific dynamic fee components.

Appendix D

Glossary

TERM	DEFINITION
Arbitrum	An Ethereum Layer 2 scaling network that provides faster, cheaper transactions while inheriting Ethereum's security guarantees.
Conviction Tier	A classification of signal strength (high, medium, research) that determines position sizing, leverage, and priority.
Epoch	A settlement cycle (typically daily) during which withdrawal requests are collected and processed at the finalization NAV.
Max Drawdown	The largest peak-to-trough decline in portfolio value, measuring the worst-case loss experienced during a period.
Mean Reversion	A statistical phenomenon where prices that move significantly away from their average tend to return toward it. The core trading thesis of REGIME.fi.
NAV (Net Asset Value)	The per-share value of the fund, calculated as total assets minus liabilities divided by outstanding shares.
Perpetual Exchange	A decentralized trading venue that offers perpetual futures contracts — leveraged positions with no expiry date.
Profit Factor	The ratio of gross profits to gross losses. A profit factor above 1.0 indicates a profitable strategy; above 1.5 is considered strong.
Share Price	Synonymous with NAV. The current USDC value of one fund share.
USDC	A US dollar-pegged stablecoin. REGIME.fi is denominated in USDC — all deposits, trades, and withdrawals use USDC.
Win Rate	The percentage of closed trades that resulted in a profit. Used alongside profit factor to evaluate strategy performance.

REGIME.fi

Autonomous Trading Protocol on Arbitrum

Dashboard GitHub

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